

The interplay of epigenetic marks during stem cell differentiation and development

Yaser Atlasi and Hendrik G. Stunnenberg

Abstract | Chromatin, the template for epigenetic regulation, is a highly dynamic entity that is constantly reshaped during early development and differentiation. Epigenetic modification of chromatin provides the necessary plasticity for cells to respond to environmental and positional cues, and enables the maintenance of acquired information without changing the DNA sequence. The mechanisms involve, among others, chemical modifications of chromatin, changes in chromatin constituents and reconfiguration of chromatin interactions and 3D structure. New advances in genome-wide technologies have paved the way towards an integrative view of epigenome dynamics during cell state transitions, and recent findings in embryonic stem cells highlight how the interplay between different epigenetic layers reshapes the transcriptional landscape.

Genomic imprints

Epigenetic markings that regulate allelic gene expression according to the parent of origin.

Epigenome

Instructive chromatin information that is not stored in the genetic code (DNA) and can be inherited by cell progeny.

Primordial germ cells

(PGCs). Embryonic cells in the gonads that give rise to gametes.

In eukaryotes, DNA is wrapped around histone octamers and is 'decorated' with structural and regulatory proteins to form the nucleoprotein chromatin structure. During development, epigenetic information (that is, extra (epi) information that is not propagated in the DNA sequence) is acquired in a progressive manner in the form of chemical modifications of chromatin, alterations in chromatin constituents and changes in the 3D chromatin organization. These changes regulate the transcriptional programme during lineage commitment or function as genomic imprints that distinguish the parental origin of genomic loci. Dynamic regulation of the epigenome underlies cellular plasticity and provides a heritable response to environmental and developmental cues. The different layers of epigenetic information are closely interconnected, and the interplay between these regulatory events and their combinatorial effects continues to be subject to extensive investigation. Epigenetic deregulation is directly linked to a wide spectrum of diseases ranging from developmental disorders associated with aberrant genetic imprinting to various cancers that have defects in protein complexes involved in histone or DNA modifications^{1,2}. The fact that epigenetic modifications are, in principle, reversible renders epigenetic regulation amenable to pharmacological intervention².

Multiple waves of epigenetic (re)setting take place during development (BOX 1). The first wave occurs after fertilization: both the maternal and paternal genomes undergo extensive epigenetic reprogramming to convert the epigenomes of the differentiated gametes to the

epigenome of the zygote, which leads to zygotic gene activation (ZGA)³. The next wave in this gradual epigenetic resetting occurs during blastocyst formation: DNA methylation is erased; histone modifications are gained, lost or redistributed; both X chromosomes are reactivated in female cells; and transcription factors are induced that increase chromatin accessibility^{4,5} (BOX 2). After implantation, another major wave of epigenetic programming takes place, in which chromatin accessibility becomes progressively restricted, deposition of repressive epigenetic marks (such as trimethylation of histone H3 at lysine 27 (H3K27me3) and DNA methylation) increases and one X chromosome is inactivated in female cells^{5–8}. A few cells will undergo an additional round of post-implantation epigenetic resetting to form the primordial germ cells (PGCs)⁹.

The process of epigenetic (re)programming has not been captured in its entirety *in vitro*; however, accumulating evidence suggests that transitions between different states of embryonic stem cells (ESCs) recapitulate to a large extent the epigenetic dynamics observed in the developing embryo^{5,10,11}. Indeed, the different states that have been captured *in vitro* provide a gradient of pluripotency that resembles different stages of embryonic development (BOX 1): naive ESCs cultured in serum-supplemented medium or in defined medium (2i medium) resemble the pre-implantation epiblast^{5,6,10}, and primed epiblast-derived stem cells (EpiSCs) resemble the post-implantation embryo^{5,6,10}. Furthermore, the different ESC states are interconvertible and provide an

Department of Molecular Biology, Faculty of Science, Radboud University, 6525 GA Nijmegen, The Netherlands.

Correspondence to H.G.S. h.stunnenberg@ncmls.ru.nl

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Pluripotency

The potential of self-renewing stem cells to give rise to all cell types of all three germ layers (but not to extra-embryonic trophoblast cells). This state is detected in early embryos and in *in vitro*-cultured mouse and human embryonic stem cells (ESCs) as well as mouse epiblast-derived stem cells (EpiSCs) and epiblast-derived stem cell-like cells (EpiSCLCs).

experimental model to elucidate the temporal epigenetic mechanisms involved in pre-implantation and post-implantation embryonic development^{6,10,12,13}. Some of the well-described epigenetic differences between naive and primed ESC states include, among others, an open chromatin conformation together with reduced DNA methylation and reduced H3K27me3 in naive ESCs, different transcriptional landscapes and differential enhancer H3K27 acetylation (H3K27ac)^{5,7,8,12,14,15}. Epigenetic (re) programming is not limited to ESCs: it also occurs during lineage differentiation of tissue-resident stem cells

or during dedifferentiation or transdifferentiation of mature cells to progenitor cells¹⁶. Thus, epigenetic marking of chromatin is highly dynamic and has a central role in different cellular transitions.

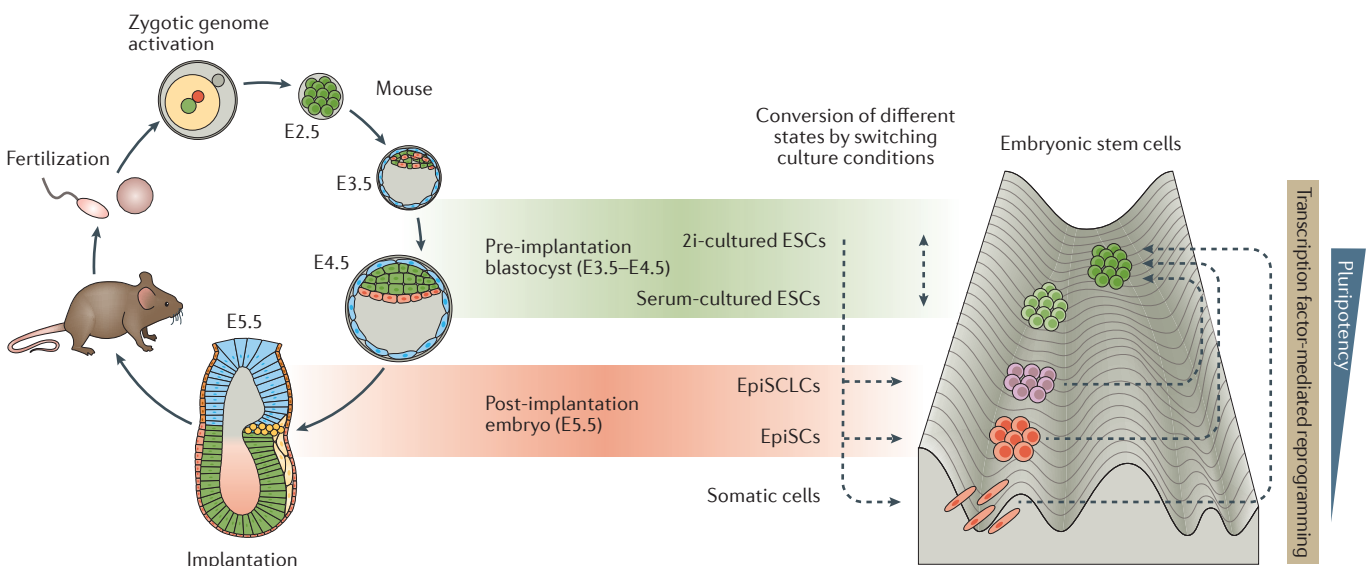
In this Review, we describe the dynamics and the relative contribution of key epigenetic regulatory events during development and stem cell differentiation. We then discuss the crosstalk between the different epigenetic marks and how their interplay controls gene expression in different cellular transitions, such as embryonic stem cell differentiation and lineage commitment in adult tissues.

Box 1 | Stages of embryonic development and the different states of pluripotency

After fertilization, the zygote undergoes consecutive rounds of cleavage to generate the blastocyst (~embryonic day 3.5 (E3.5) in mice and ~E5–E6 in humans), which consists of two major cell types: the pluripotent inner cell mass (ICM) cells that give rise to all cell lineages and the trophoblast cells that form the extra-embryonic placenta⁴ (see the figure). At later blastocyst stages (~E4.5 in mice and ~E6–E7 in humans), ICM cells undergo a second round of cell commitment to generate the pluripotent epiblast cells and the overlying hypoblast layer⁴. The E4.5 epiblast cells in mice represent the ground-state pluripotency, which is free from many epigenetic constraints that are present in the ICM or in post-implantation epiblast and is characterized by two active X chromosomes in female cells^{6,12,193}. After implantation (~E5.5 in mice and ~E8–E9 in humans), epiblast cells undergo a strong wave of epigenetic programming that includes, among others, global DNA methylation, inactivation of one X chromosome in female cells and increased deposition of repressive histone marks. These alterations ultimately restrict the differentiation potential of epiblast cells^{5,6,194}, which are therefore referred to as primed-state pluripotent cells. During post-implantation development, primed epiblast cells exit pluripotency and undergo lineage commitment to generate different somatic cell types^{5,10,194}. Pluripotency, encompassing naive and primed states, is therefore a gradient of transition states that exist in a short time-window (about 3–4 days in mice and 8–9 days in humans) at an early stage of embryonic development and before lineage specification. This pluripotent state can be captured *in vitro* in the form of different embryonic stem cell (ESC) states (see the figure).

Conventional ESCs that are derived from mouse pre-implantation embryos (or via somatic cell reprogramming) are maintained in culture conditions supplemented with serum and leukaemia-inhibitory factor (LIF). These serum ESCs are heterogeneous and have transcriptional and epigenetic differences (such as DNA methylation) compared with cells in the

pre-implantation embryo^{12,13,175}. By contrast, mouse ESCs cultured in serum-free medium supplemented with glycogen synthase kinase 3 (GSK3) and MAP/ERK kinase (MEK) inhibitors (known as 2i medium), have epigenetic (for example, DNA methylation levels) and transcriptional profiles that best resemble cells in the pre-implantation epiblast and are referred to as ground-state ESCs^{12,13,63}. Serum and 2i-cultured ESCs represent two types of 'naive' pluripotency, as both of these stem cell states contribute to chimaera formation when injected into a blastocyst or when used in tetraploid complementation assays^{195–198}. Primed-state pluripotency has also been captured *in vitro* as cultured epiblast-derived stem cells (EpiSCs) or epiblast-derived stem cell-like cells (EpiSCLCs), which resemble the early and late stage post-implantation epiblast cells, respectively^{7,8}. EpiSCs and EpiSCLCs are devoid of many naive stem cell markers, do not tolerate complete loss of DNA methylation and do not contribute to chimaera formation⁵. Conventional human ESCs have also been derived from pre-implantation blastocysts (or by somatic cell reprogramming) but show differences in epigenetic makeup compared with serum and 2i-cultured mouse ESCs, such as inactivation of one X chromosome in female cells and dependency on DNA methylation and fibroblast growth factor 2 (FGF2) signalling^{5,199}. Instead, conventional human ESCs are more similar to mouse EpiSCs and show features of primed-state pluripotency^{5,199}. Recent efforts have led to the establishment of different types of human ESCs with various features of naive-state pluripotency^{15,200–203}. For ethical reasons, human ESCs are not derived from post-implantation epiblasts. Pluripotent cells have also been established from rodent primordial germ cells (PGCs); the derived cells are termed embryonic germ cells (EGCs) and show many similarities to ground-state ESCs^{204,205}. Thus, pluripotency represents a gradient of transient states, some of which have been captured *in vitro* using specific culture conditions.



Box 2 | DNA accessibility and transcription factor binding

Chromatin accessibility represents transcription factor-binding sites, many of which are located at enhancers, and can be measured using techniques such as the DNase I hypersensitivity (DHS) assay or assay for transposase-accessible chromatin followed by sequencing (ATAC-seq)^{168,206,207}. These chromatin accessibility assays provide a robust approach to identify putative regulatory elements as well as under- or over-represented sequence motifs — potential transcription factor-binding sites — without prior knowledge of the respective transcription factor and its cis-acting regulatory sequence. By contrast, chromatin immunoprecipitation followed by sequencing (ChIP-seq) approaches²⁰⁸ examine the genome-wide binding of transcription factors (or histone marks) and therefore can be used to validate and extend the results obtained from chromatin-accessibility or motif-discovery approaches. Similarly, sequential immunoprecipitation by two different antibodies (re-ChIP)²⁰⁹ or its recent co-ChIP improvement⁶⁴ (based on sequential immunoprecipitation and barcoding of the chromatin) are used to address the colocalization of different histone marks or transcription factor binding on the same allele.

The chromatin-accessibility landscape is cell type-specific. For example, in the haematopoietic system, chromatin accessibility at enhancers is shown to be highly dynamic and can be used to classify the different cell lineages more accurately than mRNA expression analysis²⁰⁹. Importantly, recent findings indicate a progressive restriction of the DHS landscape during development; mapping DHS sites in human embryonic stem cells (ESCs) revealed a larger number of DHS sites in ESCs compared with somatic cells¹⁷. Similar observations were made in adult stem cells during the specialization of haematopoietic progenitor cells to differentiated B cells and T cells¹⁷. These observations suggest a more open or plastic chromatin structure in stem cells compared with their differentiated progeny. Interestingly, much of the ESC-specific DHS landscape is reactivated in cancer cells, leading to substantial similarity between the DHS landscape of cancer cells and ESCs¹⁷.

Epigenetic dynamics in ESCs

The chromatin of pluripotent stem cells has a unique epigenetic plasticity that enables cells to undergo a wide range of lineage specifications. Pluripotent cells have a distinct open chromatin configuration that becomes progressively restricted during development¹⁷. Loss of chromatin accessibility (BOX 2) is accompanied by increased DNA methylation and deposition (or re-distribution) of different histone marks, which affect chromatin interactions. In this section, we discuss the dynamics of the main layers of epigenetic regulation in the early embryo and we draw parallels with *in vitro*-cultured ESCs that resemble the various stages of embryonic development.

Dynamic histone marks

Different amino acids within the globular domain or within the tails of histone proteins can be post-translationally modified. Below, we discuss some of the better-described histone modifications, which have been studied in a range of experimental models and have been linked to gene activation or gene repression.

H3K4 methylation and its association with transcriptional activation. So far, H3K4me3 has been detected at promoters and in low levels at distal regulatory elements in many eukaryotes, ranging from yeast to human¹⁸. It is deposited by protein complexes that include members of the histone methyltransferase family and that were initially discovered in the yeast *Saccharomyces cerevisiae*, where they are called complex proteins-associated with SET1 (COMPASS) complexes^{19,20}; in mammals, they are termed COMPASS-like complexes. Animal models depleted for the core components of COMPASS-like complexes

— for example, SET1 (also known as SETD1A), mixed lineage leukaemia protein 1 (MLL1; also known as KMT2A) and MLL2 — survive pre-implantation development but die after gastrulation^{21,22}. Depleting mouse ESCs of H3K4me3 writers (that is, the COMPASS-like proteins) generates a range of phenotypes, including reduced self-renewal^{23,24} and impaired differentiation²⁵, implying that COMPASS-like proteins regulate distinct subsets of genes and/or are involved in biological functions other than depositing the H3K4me3 mark.

H3K4me3 is deposited in broad domains in the oocyte, and becomes restricted to transcription start sites after fertilization and during the major wave of zygotic genome activation^{26–28} (FIG. 1). The regulation of H3K4me3 seems to be less dynamic than other histone marks: its genome-wide level remains relatively stable in different ESC states^{12,29}, during the differentiation of human ESCs³⁰ and between different cell lines³¹. The roles of H3K4me3 in gene expression remain unknown. Biochemically, H3K4me3 facilitates assembly of the transcriptional machinery and mediates more efficient induction of gene expression in response to environmental cues^{32,33}. However, in *S. cerevisiae*, there is little evidence that H3K4me3 mediates specific gene activation²⁰. Furthermore, in vertebrates, the deposition of H3K4me3 does not predict the level of gene expression; instead, it might precede transcription, for example, during the maternal-to-zygotic transition in *Xenopus laevis*³⁴, or it might even mark a subset of inactive genes³³. Thus, although a general correlation between H3K4me3 and gene expression has been observed³⁵, H3K4me3 deposition is not an accurate discriminator of the cellular state or transcriptional activity.

The monomethylation (H3K4me1) or dimethylation (H3K4me2) marks on H3K4 have broader patterns compared with H3K4me3 and are mainly detected at intergenic sites^{31,36,37}. The role and dynamics of H3K4me1 and H3K4me2 have been described in distal regulatory elements, where H3K4me1 and H3K4me2 function as general enhancer marks and demarcate cell-specific enhancer repertoires³⁸. H3K4me1 is mainly deposited by protein complexes containing MLL3 and MLL4 (REF. 18) (FIG. 1). In mice, MLL3 is dispensable for embryonic development (mice die around the time of birth), whereas loss of MLL4 is embryonic lethal³⁶. These observations suggest a partial redundancy of MLL3 and MLL4 in mouse development. The loss of *Mill3* and *Mill4* function does not affect the self-renewal of mouse ESCs or of mouse embryonic fibroblasts but strongly suppresses their potential for differentiation and their susceptibility to somatic cell reprogramming, respectively³⁹. Complementary to these observations, mouse ESCs and mouse haematopoietic stem cells (HSCs) that lack the lysine demethylase 1 (*Lsd1*; also known as *Kdm1*) gene, which encodes an eraser of H3K4me1, continue to self-renew but fail to undergo proper differentiation because of defective decommissioning of stem cell-specific enhancers^{40,41}. Together, these data suggest that the replenishment of H3K4me1 is not required for maintaining cellular identity under steady-state conditions when enhancers are established, but that H3K4me1 (and

2i medium

A serum-free and defined culture medium that is supplemented with MAPK/ERK kinase (MEK) and glycogen synthase kinase 3 (GSK3) inhibitors.

Epiblast-derived stem cells (EpiSCs). *In vitro*-cultured stem cells that are derived from the post-implantation epiblast.

Tetraploid complementation assays

A technique that combines diploid embryonic stem cells (ESCs) and tetraploid embryos to examine the pluripotency potential of ESCs to generate a viable organism.

COMPASS-like complexes

Complex proteins associated with SET1 complexes are multisubunit protein complexes that are involved in mono-, di- and tri-methylation of histone H3 at lysine 4 (H3K4) and include many enzymatic proteins such as mixed lineage leukaemia protein 1 (MLL1), MLL2, MLL3 and MLL4.

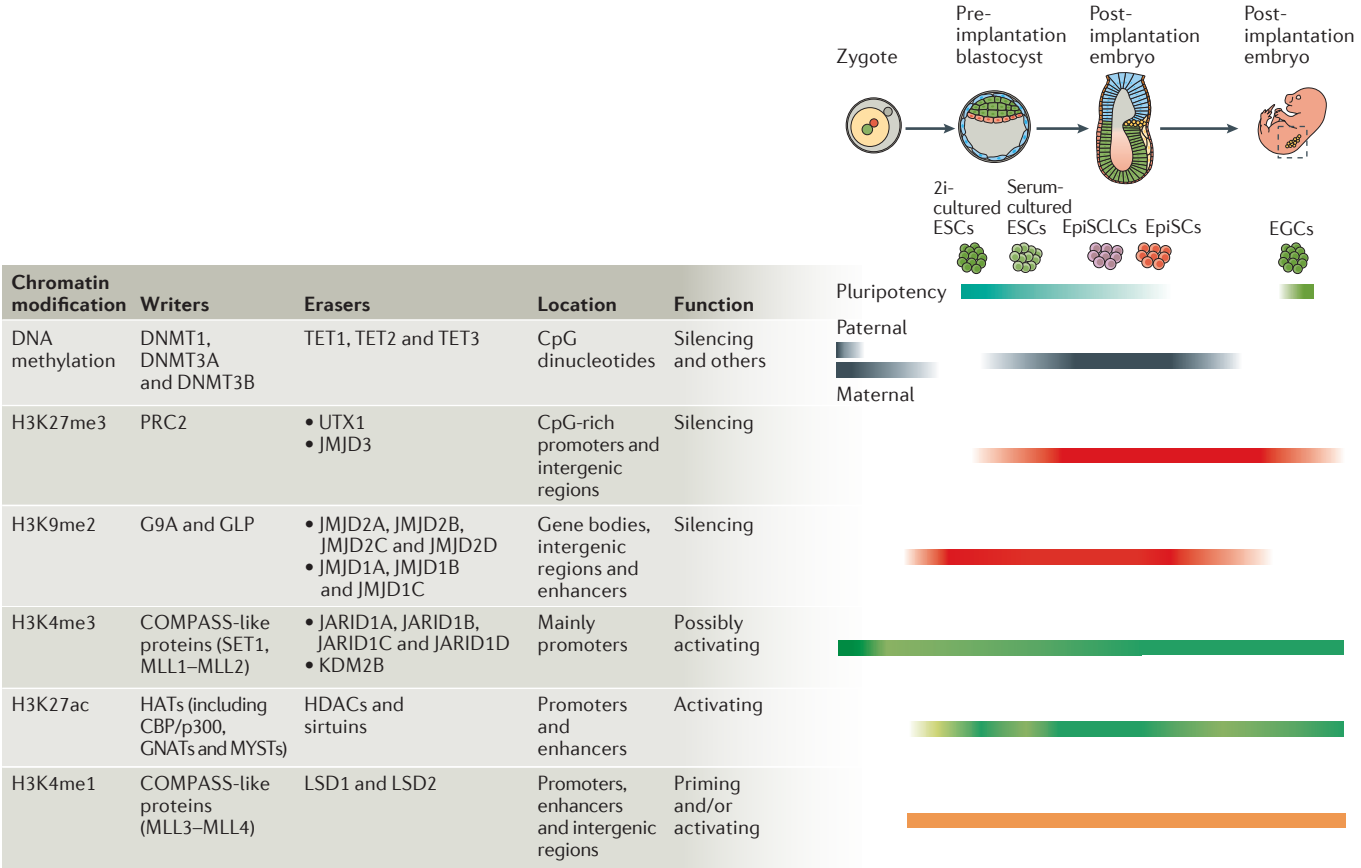


Figure 1 | Dynamics of epigenetic modifications during early mouse development and ESC differentiation. The main epigenetic marks discussed in this paper are shown. The location, function and the main protein complexes that deposit or remove these epigenetic marks (writers and erasers) are listed. DNA methylation is erased in the paternal and maternal genomes after fertilization and is deposited again at later developmental stages, in serum embryonic stem cells (ESCs) and in primed cells (epiblast-derived stem cells (EpiSCs) and epiblast-derived stem cell-like cells (EpiSCLCs)). DNA methylation is reduced during primordial germ cell (PGC) development. Similarly, the silencing histone marks (for example, trimethylation of histone H3 at lysine 27 (H3K27me3) and H3K9me2) increase post-implantation and are reduced in 2i-cultured ESCs and PGCs. H3K4me3 is present in broad domains in the oocyte but is restricted to transcription start sites after fertilization. CBP, CREB-binding protein; COMPASS-like proteins, complex proteins associated with SET1; DNMT, DNA methyltransferase; EGCs, embryonic germ cells; GLP, G9A-like protein 1; GNATs, Gcn5-related N-acetyltransferases; HATs, histone acetyltransferases; HDACs, histone deacetylases; JARID, Jumonji, AT-rich interactive domain (JARID1A–JARID1D are also known as KDM5A–KDM5D); JMJD, Jumonji domain-containing protein (JMJD3 is also known as KDM6B; JMJD2A–JMJD2D are also known as KDM4A–KDM4D; JMJD1A–JMJD1C are also known as KDM3A–KDM3C); KDM, lysine-specific demethylase; LSD, lysine demethylase (LSD1 and LSD2 are also known as KDM1A and KDM1B, respectively); MLL, mixed lineage leukaemia protein; MYSTs, MOZ, YBF2/SAS3, SAS2 and TIP60 proteins; PRC2, Polycomb repressive complex 2; TET, ten–eleven translocation; UTX1, ubiquitously transcribed TPR on X (also known as KDM6A).

Writers
Enzyme (complexes) depositing chemical modifications in chromatin. These proteins include, among others, histone modifiers (for example, histone acetyltransferase or histone methylases) and DNA modifiers (such DNA methyltransferases).

Eraser
An enzyme that removes chemical modifications from chromatin, such as histone deacetylases or histone demethylases.

the MLL3–MLL4 protein complex) becomes important for establishing *de novo* (lineage-specific) enhancers when cells are triggered for cellular transitions, such as upon cell differentiation.

H3K27 acetylation is associated with activation of gene expression. H3K27 acetylation is deposited at active regulatory elements (predominantly promoters and enhancers) by different histone acetyltransferases (HATs), such as the CREB-binding and p300 (CBP/p300) proteins, and is strongly and positively correlated with gene expression^{37,42,43} (FIG. 1). H3K27ac is linked to the activation of enhancers that are primed (marked by H3K4me1) or poised (marked by H3K4me1 and

H3K27me3) and in response to environmental stimuli or during different developmental stages^{37,44}. The H3K4me1 and H3K27ac marks partially overlap; in mouse ESCs, about 50% of H3K4me1 enhancers are co-marked by H3K27ac³⁷. Conversely, the majority of H3K27ac-marked enhancers also have H3K4me1, with approximately one-third of all H3K27ac-marked enhancers lacking H3K4me1 (REF. 37). About 70% of enhancers that are marked with H3K27ac are active enhancers — that is, they boost transcription *in vivo*⁴⁵. A subset of active enhancers has been identified in mouse ESCs that lack H3K27ac but are marked by H3K122ac and/or H3K64ac, H3K4me1 and enhancer RNAs (eRNAs) (BOX 3), and such enhancers have been shown

Readers

Proteins that recognize and bind to chemical modifications on chromatin, such as bromodomain-containing proteins that recognize acetylated histones or other proteins.

Inner cell mass

(ICM). Cells in the pre-implantation embryo (blastocyst) that will eventually give rise to different cell types in the embryo.

Polycomb repressive complex 2

(PRC2). Polycomb repressive complexes are multisubunit protein complexes that are writers of histone marks. These include PRC1, which is responsible for mono-ubiquitylation of histone H2A at lysine 119 (H2AK119Ub1), and PRC2, which is responsible for trimethylation of histone H3 at lysine 27 (H3K27me3).

to activate gene expression in cell culture and *in vivo*⁴⁶. Thus, although H3K27ac marks reflect enhancer activity, enhancers that lack H3K27ac are not necessarily inactive; other histone marks can carry out the activating role of H3K27ac.

Expression of histone deacetylase 1 (HDAC1), which is the major deacetylase in the pre-implantation mouse embryo, decreases in the blastocyst. Knockdown of *Hdac1* delays blastocyst formation and induces lethality after implantation⁴⁷, and *Hdac1*-null ESCs can be maintained *in vitro* but show lower levels of proliferation^{48,49}. These and other observations^{50,51} suggest a potential link between histone acetylation and maintenance of pluripotency in early embryos and in cultured ESCs. Depleting the mouse bromo- and extra-terminal domain (BET) proteins, which are readers of acetylated histones, also leads to embryonic lethality: bromodomain-containing protein 4 (*Brd4*)-mutant embryos die shortly after implantation and blastocyst explants of *Brd4*-mutant mice cannot establish inner cell mass (ICM) identity *in vitro*⁵². In mouse ESCs, inhibition of BRD4 using the JQ1 small molecule inhibitor decreases histone acetylation and leads to reduced self-renewal and increased differentiation⁵¹, further supporting a role for H3K27ac in maintaining ESC self-renewal. Taken together, H3K27ac is highly dynamic in different cellular states and its deposition is correlated with enhancer activation and transcriptional activity.

Box 3 | Enhancer RNAs and bidirectional transcription

In addition to chromatin modifications that represent a major layer of epigenetic regulation, other mechanisms such as expression of non-coding RNAs can control chromatin modification and organization. Among these non-coding RNAs are the small (<2 kb) bidirectional RNAs termed enhancer RNAs (eRNAs) that are detected at distal regulatory elements²¹⁰. These unstable RNAs are usually 5'-capped and non-polyadenylated, rarely undergo splicing, are retained in the nucleus, and their chromatin templates are not marked by trimethylation of histone H3 at lysine 36 (H3K36me3)^{38,211}. Enhancer transcription (that is, eRNA production) marks only a subset of cis-regulatory elements^{211,212}. The Functional Annotation of the Mammalian Genome (FANTOM) consortium sequenced 25–27 nucleotides from the 5' end of RNAs to detect widespread transcription of bidirectional RNAs in the genome²¹¹. Using this approach, ~11% of regions marked by H3K4me1 and H3K27 acetylation (H3K27ac), and only 4% of DNase I hypersensitivity (DHS) sites showed eRNA expression. However, of those enhancers that express eRNAs, 71% are marked by H3K4me1 and H3K27ac, and 87% overlapped DHS sites. These findings suggest that eRNA transcription marks a subset of enhancers that show strong activity^{211,212}.

Emerging data suggest that eRNAs are functionally important for enhancer activity and for regulating gene expression, and their functions include RNA polymerase II (RNAPII) loading, RNAPII pause release and recruitment of different transcription factors^{213,214}. eRNAs have also been functionally linked to enhancer–promoter looping and 3D chromatin organization^{212,215–218}, for example, through interaction with the cohesin²¹⁹ and the Mediator complexes²²⁰. Knockdown of eRNA expression in human and mouse cells reduces the expression of their corresponding target genes and, in a few examples, attenuates chromatin looping^{219–221}. Similarly, depleting Integrator, a multisubunit complex required for the processing of small nuclear RNA and eRNAs, impairs eRNA processing and reduces enhancer–promoter looping²²². eRNAs were also shown to increase H3K27 acetylation at enhancers by stimulating the acetyltransferase activity of CREB-binding protein (CBP), a major histone acetyltransferase (HAT) protein, and to increase the transcription of target genes²²³. Collectively, eRNAs play an essential part in enhancer function by influencing the chemical modification of chromatin, mediating spatial chromatin interactions and regulating gene transcription.

H3K27 trimethylation is associated with repression of gene expression. H3K27me3 is associated with gene silencing and is mainly deposited at CpG-rich promoters but can also be detected at intergenic regions^{53,54} (FIG. 1). H3K27me3 is often observed at DNA regions that are unmethylated or that have low levels of DNA methylation^{55,56}, suggesting a potential antagonism between H3K27me and DNA methylation. Large regions (100 kb and more) of the homeobox (*HOX*) loci (which encode genes that control the body plan) are marked with H3K27me3, which is readily detectable on these and other modified regions in the oocyte, 2-cell embryo and early-stage blastocyst^{27,53,57}. H3K27me3 is also prevalent on the heterochromatin of the inactivated X chromosome, implying a functional role for H3K27me3 in maintenance of X inactivation in female cells⁵⁸.

In mouse embryos, H3K27me3 is gradually gained from the 2-cell stage to the 16-cell stage, followed by a particularly marked increase during the transition from the morula to the early blastocyst stage²⁷. In serum-cultured mouse ESCs, H3K27me3 decorates promoters of developmental genes^{59,60}. Depleting components of Polycomb repressive complex 2 (PRC2; the writer of the H3K27me3 mark) in these cells leads to activation of PRC2 target genes and increased spontaneous cell differentiation^{61,62}. By contrast, H3K27me3 is strongly reduced at these promoters in 2i-cultured ESCs⁶³ (BOX 1; FIG. 1), but surprisingly without causing transcriptional activation^{12,64}. This observation suggests that either unknown mechanisms can substitute for H3K27me3 function in gene silencing or that 2i-cultured cells do not express the transcription factors required for transcriptional activation of these promoters. PRC2 is dispensable for mouse ESC self-renewal^{65–67}, and depleting PRC2 components does not affect pre-implantation development, although embryos lacking PRC2 die shortly after gastrulation^{68–70}. However, knockdown of PRC2 components in mouse EpiSCs, which emerge in the post-implantation embryo, reduces cellular viability and self-renewal⁷¹. Furthermore, PRC2 is crucial for homeostasis of mouse adult stem cells by suppressing lineage commitment genes, and depleting H3K27me3 impairs proper stem cell differentiation^{72–74}.

Taken together, these observations suggest that H3K27me3 is dispensable for transcriptional repression in naive mouse ESCs^{12,75,76} but is essential at later developmental stages, particularly in post-implantation mouse embryos. Accumulating data indicate that deposition of H3K27me3 marks might occur after gene silencing, as transcriptional repression results in the recruitment of the PRC2 complex⁷⁵. Moreover, mouse ESCs lacking PRC2 show dysregulated transcriptional repression only after long-term differentiation, which suggests that H3K27me3 is required for maintaining, rather than initiating, gene silencing⁷⁵.

H3K9 dimethylation is associated with repression of gene expression. H3K9me2 is widely distributed on chromatin in large domains that can extend to 4 Mb in size⁷⁷. H3K9me2 is strongly associated with gene silencing and, in contrast to H3K27me3, it is deposited on gene bodies and in many intergenic regions^{77,78} (FIG. 1). A subset

of enhancers gains H3K9me2 and becomes repressed in the post-implantation embryo or in mEpiSCs^{78,79}. Subsequently, many intergenic sites lose H3K9me2 during PGC reprogramming^{29,80}. These two observations suggest a role for this mark in decommissioning enhancers associated with the naive state⁷⁹.

H3K9me2 is deposited by a protein complex that includes G9A (also known as EHMT2) and G9A-like protein 1 (GLP; also known as EHMT1). Depleting mouse embryos of either of these proteins leads to embryonic lethality at approximately E9.5 (REFS 81,82), which indicates that H3K9me2 is required for post-implantation development. However, G9A-null mouse ESCs can be maintained in culture, suggesting that H3K9me2 has a minimal role in self-renewing ESCs⁸¹. The role of H3K9me2 in mouse ESC differentiation remains controversial. Initial observations indicated that levels of H3K9me2 increased during differentiation⁷⁷; however, other studies have questioned this finding and described an overall conserved pattern of H3K9me2, with only a small increase affecting specific sites (5% of regions) in differentiated cells^{29,78}. Thus, a focal increase in H3K9me2 deposition seems to silence enhancers associated with the naive state, and erasing H3K9me2 is linked to the transition from a more committed-state to naive-state pluripotency. Similarly, in human HSCs, deposition of H3K9me2 increases during lineage commitment, and inhibiting G9A activity by small-molecule inhibitors facilitates *in vitro* expansion of HSCs⁸³.

Dynamic DNA modifications

DNA methylation is a crucial player in epigenetic regulation and is so far the best-studied epigenetic modification in mammals. A first wave of DNA demethylation takes place immediately after fertilization, during which the paternal genome is demethylated more rapidly than the maternal genome^{84,85}. In humans, this wave of demethylation is completed at the 2-cell embryo stage^{86,87}. This process is more gradual in mice and continues until the blastocyst stage⁸⁸ (FIG. 1); DNA methylation reaches its lowest level in blastocyst-stage mouse embryos and sharply increases post-implantation. A second wave of DNA demethylation takes place in two phases in post-implantation PGCs¹¹ and is closely linked to PGC reprogramming and induction of the pluripotency programme. The first phase of demethylation occurs by replication-dependent dilution of methylated DNA and takes place during migration of PGC precursors. The second phase involves active demethylation and occurs after the entry of PGCs into the gonads^{89,90}. Below, we briefly discuss how DNA methylation is deposited and erased in different cellular contexts.

Deposition of DNA methylation. DNA methylation takes place at CpG dinucleotides and is mediated by members of the DNA methyltransferase (DNMT) family of enzymes that include DNMT1, DNMT3A, DNMT3B and DNMT3L^{11,91}. DNMT1 is involved in maintenance of DNA methylation; it recognizes hemimethylated DNA and restores it to the fully methylated state after DNA replication⁹². It is closely associated with the E3 ubiquitin protein ligase UHRF1, and together, they are recruited

to hemimethylated sites and regions that are marked by H3K9 methylation, which indicates a connection between DNA methylation and H3K9 methylation^{93–96}. By contrast, DNMT3A and DNMT3B (in conjunction with DNMT3L, which lacks methyltransferase activity) do not show preference for hemimethylated DNA and instead mediate *de novo* DNA methylation⁹¹. Levels of the three DNMTs (DNMT3A, DNMT3B and DNMT3L) increase markedly in post-implantation epiblasts and in *in vitro*-cultured EpiSCs, and the activity of these proteins mediates the genome-wide remethylation of DNA¹¹. Depleting DNMT3A and DNMT3B enzymes in mouse ESCs leads to a gradual reduction in DNA methylation in long-term cultures, indicating that DNMT1 alone is not sufficient for long-term maintenance of DNA methylation^{97,98}. A previously annotated pseudogene in rodents was recently reported to encode a protein with *de novo* DNA methylation activity and was termed *Dnmt3c*⁹⁹. DNMT3C specifically functions during spermatogenesis to mediate methylation and silencing of retrotransposon elements⁹⁹. Accordingly, male animals depleted for DNMT3C are sterile, which supports a specific role for this protein during male germ cell development⁹⁹.

Erasing DNA methylation and the role of TET proteins.

DNA methylation is a reversible process. DNA demethylation is achieved either by passive replication-dependent dilution of methylation or by active enzymatic demethylation¹⁰⁰. In the case of active enzymatic demethylation, 5-methylcytosine (5mC) undergoes a stepwise chemical oxidation by the methylcytosine dioxygenase TET proteins (TET1, TET2 and TET3) that ultimately leads to removal of the mark⁹⁰. 5-Hydroxymethylcytosine (5hmC) is the best-studied and first intermediate of this enzymatic cytosine demethylation. 5hmC is mainly found at enhancers, and depleting the three TET enzymes in mouse ESCs results in increased DNA methylation at 15–25% of all enhancers¹⁰¹. *Tet1* and *Tet2* are highly expressed in mouse ESCs and are downregulated on cell differentiation¹⁰², whereas *Tet3* mainly demethylates the sperm genome after fertilization⁸⁴. Mice lacking both TET1 and TET2 are viable and overtly healthy¹⁰³. Mouse ESCs that lack all three TET proteins maintain self-renewal, despite their complete lack of 5hmC; however, these cells show differentiation defects and fail to contribute to chimeric embryos, which indicates a role for TET proteins in ESC differentiation¹⁰⁴. Thus, both DNA methylation and TET function are dispensable for mouse ESC self-renewal but are required for exit from pluripotency and for proper differentiation^{11,13,88,103–105}. The rapid post-fertilization demethylation of the sperm genome was initially explained by TET3-driven oxidation of 5mC to 5hmC^{84,85}. However, more recent findings show that depleting *Tet3* from mouse embryos prevents accumulation of 5hmC but does not influence early demethylation of the paternal genome¹⁰⁶. The existence of an alternative unknown mechanism for DNA demethylation in the early mouse zygote has therefore been suggested¹⁰⁶, but the exact role of *Tet3* and 5hmC in this process remains to be addressed. In contrast to the paternal genome, maternal DNA has been shown to lose methylation mainly via replication-dependent demethylation in mice¹⁰⁰.

5-Methylcytosine

(5mC). Methylated form of cytosine that is found in DNA and RNA molecules. DNA methyltransferase 3A (DNMT3A), DNMT3B, DNMT3C and DNMT1 are the writers of 5mC on DNA.

5-Hydroxymethylcytosine

(5hmC). Oxidized form of 5-methylcytosine that is generated by the action of TET proteins. 5hmC can be further converted into formyl and carboxyl methylcytosine, which can be removed from DNA to generate unmethylated cytosine.

DNA methylation reaches its lowest levels in pre-implantation embryos, PGCs and in 2i-cultured ESCs (all of which show naive-state pluripotency; see BOX 1) but strongly increases in post-implantation epiblasts, EpiSCs and in serum-cultured ESCs^{13,88–90}. Several different studies have aimed to decipher the underlying mechanisms that mediate DNA demethylation in ground-state pluripotency^{13,89,107–109}. Mathematical modelling combined with analysis of different *Dnmt*-knockout mouse ESCs suggests that in ground-state mouse ESCs (2i-cultured cells), and by inference in mouse PGCs, the overall reduction in global DNA methylation is mainly mediated by impaired maintenance of methylation and particularly by preventing DNMT1 function through reduced levels of UHRF1 and H3K9me2 (REF. 107). Depleting TET proteins in mouse ESCs did not affect the global dynamics of DNA demethylation during serum-to-2i transition¹⁰⁷, indicating a minimal role for TET enzymes in this process and suggesting that genome-wide demethylation is mainly mediated by mechanisms that impair *de novo* DNA methylation¹⁰⁷.

Taken together, DNA methylation shows a dynamic pattern of establishment and erasure in different cellular transitions. It is erased in pre-implantation development, PGCs and naive pluripotent cells (for example 2i-cultured ESCs), seemingly by passive demethylation and independently of TET-mediated 5hmC production. However, the rate of demethylation in some cellular contexts, such as demethylation of paternal and maternal genomes or demethylation in later stages of PGC development, cannot be solely explained by passive demethylation, and therefore other mechanisms involving active demethylation (both TET-dependent and TET-independent) are likely to play a part.

Dynamic 3D chromatin organization

The interphase genome is compartmentalized into topologically associated domains (TADs), which are the structural and functional units of chromatin^{110,111} (FIG. 2). Studies have shown that TADs are relatively stable between different cell types and developmental stages¹¹² and, to some extent, even between species¹¹⁰. TAD boundaries are enriched in bound CTCF (76% of all boundaries show CTCF binding), which is thought to form physical constraints that restrict the spread of epigenetic marks to neighbouring regions¹¹³ and constrains enhancer activities. Indeed, deleting CTCF-binding sites, such as those located in *HOX* clusters, results in the expansion of active chromatin into the adjacent repressed TAD during mouse ESC differentiation¹¹⁴. Thus, TADs represent an architectural framework to divide the chromatin into insulated units with respect to epigenetic spreading. Groups of TADs adopt a higher level of chromatin organization in the form of A and B compartments¹¹⁵ (FIG. 2b). A compartments are associated with active chromatin, whereas B compartments are linked to inactive chromatin. Compared with TADs, the organization of A and B compartments is more dynamic, and TADs can switch compartments, depending on the cellular state and epigenetic marking^{112,116–119}.

Studies in humans and mice indicate that interactions within TADs are also organized into densely connected loops, termed insulated neighbourhoods^{120,121} (FIG. 2c). These insulated neighbourhoods are thought to represent sub-TAD clusters of interconnected enhancer–promoter interactions that embed genes and form physically separate loops flanked by CTCF and the cohesin complex^{120,122}. Depleting the cohesin complex diminishes the number of interactions within the TADs but does not affect the overall TAD organization¹²³; thus, insulated neighbourhoods have been suggested to represent the functional units of TADs. Emerging data suggest that sub-TAD looping (which probably occurs within the insulated neighbourhoods) ranges from very stable to highly dynamic, depending on the cell type. In naive and primed human ESCs, which represent two closely related cell states, sub-TAD looping is mostly stable¹²². By contrast, the sub-TAD looping changes substantially upon differentiation of ESCs^{112,124}, during somatic cell reprogramming^{116,125–127} and in haematopoiesis^{117,118}. However, the effect of sub-TAD looping and insulated neighbourhoods on gene regulation is poorly understood; whether and how they affect gene expression and how dynamic enhancer–promoter interactions are controlled at the level of the 3D chromatin structure are areas of intense investigation.

In summary, different layers of epigenetic regulation show dynamic patterns at different stages of embryonic development and in different cellular states. Importantly, repressive epigenetic marks such as DNA methylation, H3K27me3 and H3K9me2 seem to be dispensable for naive-state pluripotency and for pre-implantation development. Deposition of repressive epigenetic marks increases post-implantation and in primed-state pluripotency, in which their roles are vital for embryonic development. By contrast, a general decrease in chromatin accessibility and/or transcription factor binding at the sub-TAD level in combination with activating histone marks, such as acetylation of histone H3, is observed during differentiation, which can restrict the pluripotency potential of stem cells.

The interplay of epigenetic hallmarks

A subject of debate has been and still is whether histone modifications are a cause or a consequence of transcriptional activity. Moreover, the crosstalk between the different epigenetic layers and whether they have synergistic, antagonistic or redundant roles in different cellular contexts remains an area of intense research. Below, we discuss the interplay between different epigenetic marks and regulatory layers during development and stem cell differentiation.

Histone marks and DNA accessibility

Enhancers marked by H3K4me1 and H3K27ac promote transcription and the loss of H3K27ac and/or H3K4me1 is linked to enhancer decommissioning^{37,40,128}. The H3K4me1 modification seems to precede H3K27ac deposition and to prepare the enhancers for further activation by H3K27ac, a concept referred to as enhancer priming^{37,129} (FIG. 3). For example, during the differentiation of human ESCs to pancreatic cells, pancreas-specific

Topologically associated domains

(TADs). TADs are densely interacting regions of the genome that represent the structural units of 3D chromatin organization in interphase cells.

CTCF

An 11-zinc-finger protein that functions as an insulator of genomic loci; it interacts as a homodimer with two CTCF motifs to mediate chromatin looping.

Cohesin complex

A multisubunit complex that holds sister chromatids together after replication. Together with CTCF, it mediates the formation of chromatin interaction loops (such as enhancer–promoter interactions). Subunits of the cohesin complex include structural maintenance of chromosomes protein 1 (SMC1), SMC3 and RAD21 proteins.

Primed-state pluripotency

A restricted state of pluripotency that exists in early post-implantation epiblasts and their *in vitro*-derived cells.

Enhancer priming

Pre-marking of enhancers with monomethylation of histone H3 at lysine 4 (H3K4me1) in progenitor cells for later activation (for example, with H3K27 acetylation (H3K27ac)) in differentiated cells.

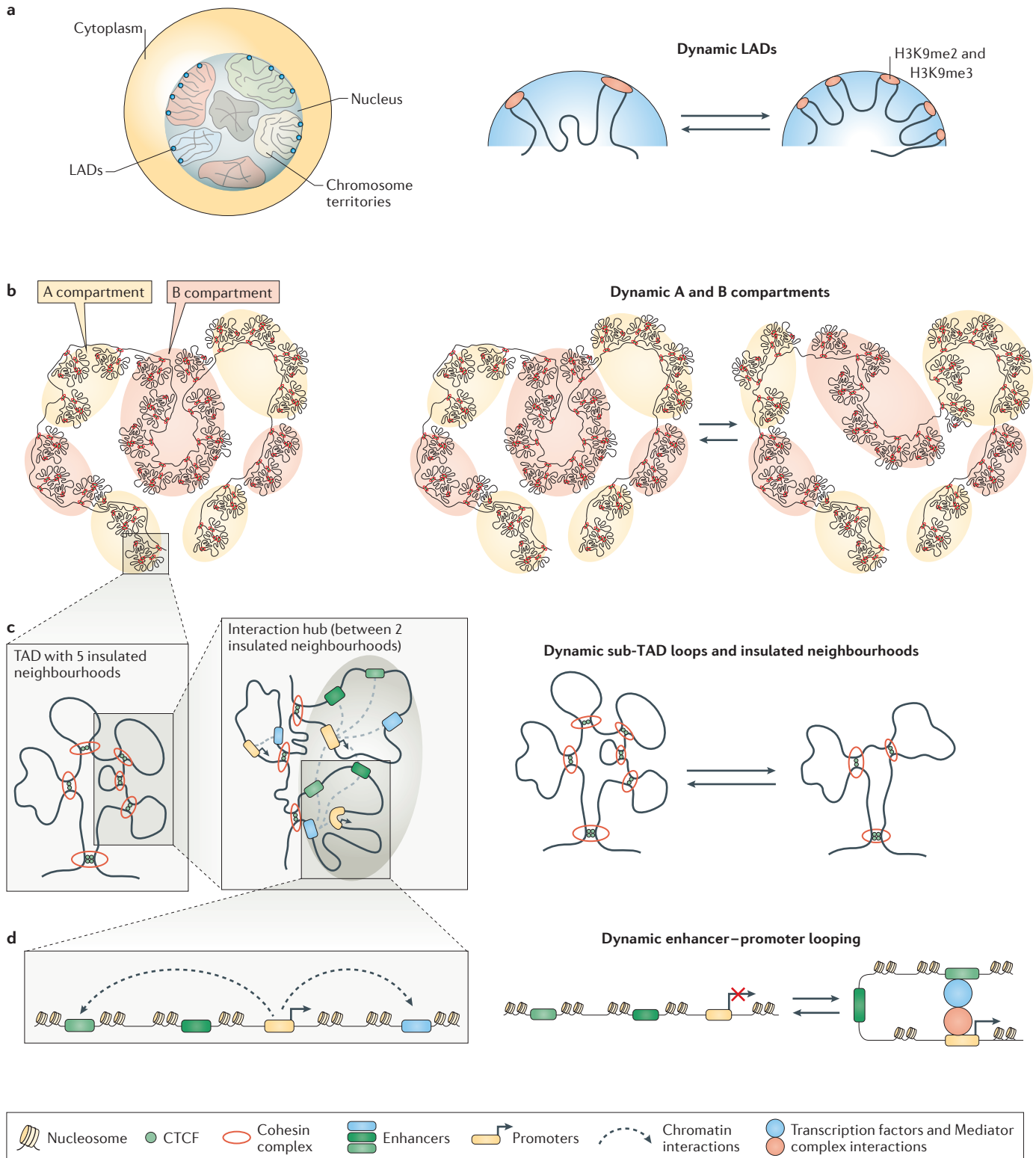


Figure 2 | 3D organization of the genome. a | Lamina-associated domains (LADs) are marked by histone H3 lysine 9 dimethylation (H3K9me2) or trimethylation (H3K9me3) and are positioned close to the nuclear membrane. **b** | Topologically associated domains (TADs) can acquire different epigenetic modifications and switch between more active compartments (A compartments) or less active compartments (B compartments). **c** | TADs are delineated by the CTCF–cohesin complex and contain several insulated neighbourhoods. Interaction hubs represent densely connected chromatin

interactions between different insulated neighbourhoods. TAD organization is mostly stable between different cell types but sub-TAD organization (such as insulated neighbourhoods and interaction hubs) change depending on the extent of differentiation and epigenetic modifications. **d** | Enhancer–promoter sub-TAD loops are mediated by different transcription factors that bring the enhancers in close proximity to the corresponding promoters. Induction or suppression of gene expression may or may not be mediated by establishing *de novo* enhancer–promoter interactions.

Pioneer transcription factor
A transcription factor that recognizes closed chromatin regions and makes these regions more accessible to binding of other transcription factors.

enhancers acquire the H3K4me1 mark and become primed in progenitor cells, before acquiring H3K27ac and becoming active in terminally differentiated cells¹³⁰.

This raises the question of how writers know where to place the mark. Most, if not all, depositing-enzyme complexes lack sequence-specific DNA-binding properties^{131,132}, so their interaction with a DNA-binding factor (or factors) is likely to direct the deposition of the initial mark. For instance, the transcription factor forkhead box protein A1 (FOXA1) was shown to be associated with enhancer priming and to function as a so-called pioneer transcription factor in differentiating intermediate cells before induction of pancreas lineages¹³⁰. Pioneer transcription factors recognize and bind to closed chromatin regions (that lack pre-existing DNase hypersensitivity (DHS) or histone marks) and initiate a sequence of events that leads to chromatin opening and to binding of other transcription factors and epigenetic modifiers¹³³. In a pancreas study, the transcription factor pancreas-duodenum homeobox protein 1 (PDX1) was shown to be necessary for H3K27ac deposition and for activation of pancreas-specific enhancers¹³⁰. In another study, naive enhancers were examined during differentiation of mouse naive ESCs to primed EpiSCs^{134,135}. In this cell state

transition, naive stage-specific enhancers lose H3K27ac and p300, a key histone acetyltransferase that acetylates H3K27, but mostly maintain the H3K4me1 mark. The transcription factor FOXD3 was shown to mediate this process by decommissioning naive enhancers and reducing H3K27ac and H3K4me2; depleting FOXD3 in naive mouse ESCs prevented enhancer decommissioning and the ESC-to-EpiSC transition^{134,135}. Later in development, inhibiting FOXD3 activity was shown to be necessary for re-activation of naive enhancers and for mouse PGC reprogramming. These and other observations^{133–136} suggest that binding of pioneer transcription factors activates enhancers by generating a permissive chromatin state that directs H3K4me1 deposition, which is in turn followed by H3K27ac marking (FIG. 3). Whether maintenance of H3K27ac at enhancers depends on the H3K4me1 mark is less clear. In mouse ESCs that lack the genes that encode MLL3 and MLL4 (which are writers of H3K4me1), only 30% of MLL4 target sites lose H3K27ac³⁹. This observation indicates that once a certain level of H3K27ac is reached, maintenance of H3K27ac is no longer dependent on the H3K4me1 mark. However, ESCs lacking MLL3 and MLL4 maintain a low 'basal' level of H3K4me1, which is deposited by other MLL family proteins³⁹. These experiments

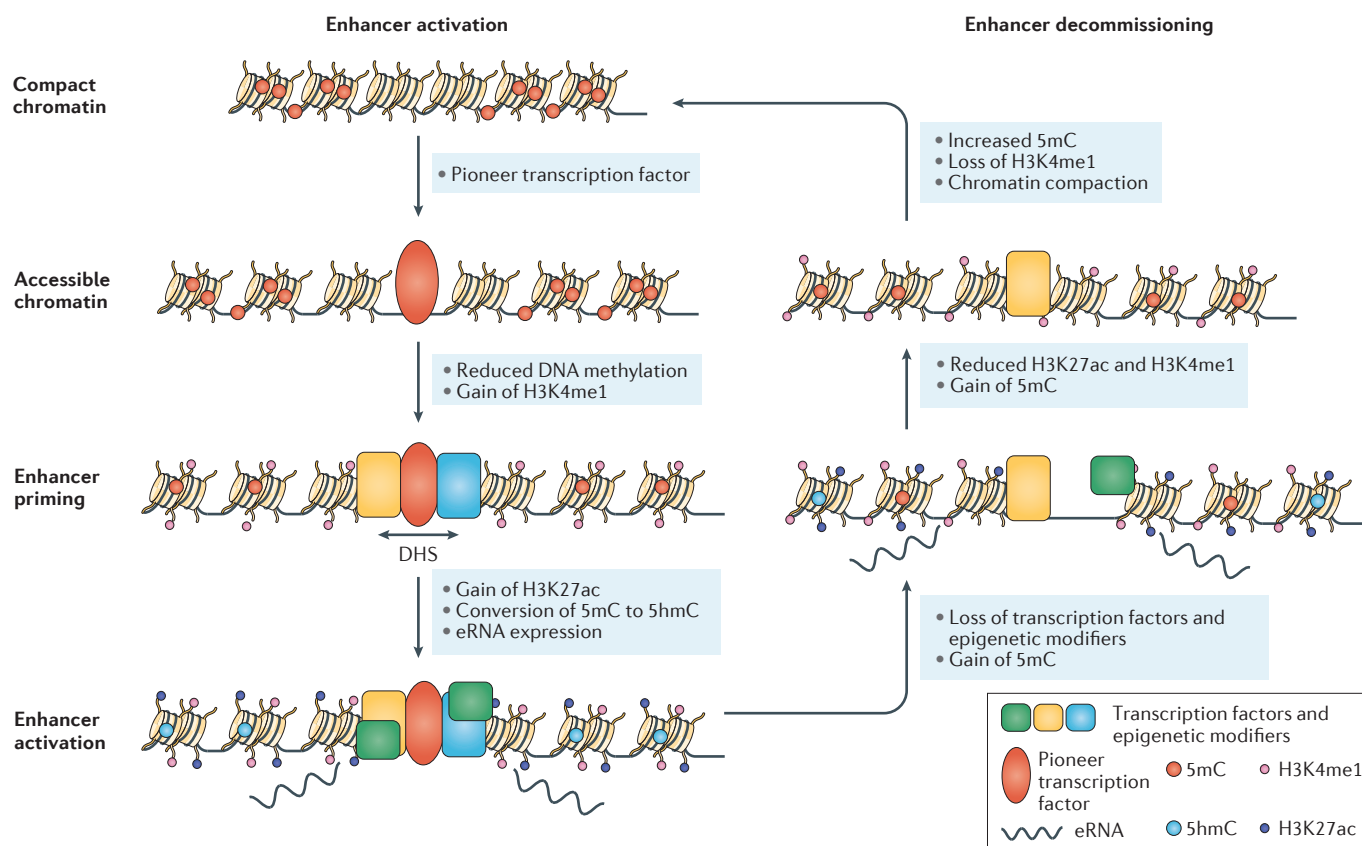


Figure 3 | Sequence of events during enhancer activation and decommissioning. Enhancer activation is initiated by binding of pioneer transcription factors that increase the chromatin accessibility and generate a permissive chromatin environment for other transcription factors and epigenetic modifiers to bind. Enhancers are often primed by the monomethylation of histone H3 at lysine 4 (H3K4me1) mark and are fully activated upon deposition of H3K27

acetylation (H3K27ac). Enhancers are decommissioned by release of transcription factors, which is accompanied by reduced chromatin accessibility, removal of the H3K4me1 and H3K27ac histone marks and finally increased DNA methylation to generate a heterochromatin state that seals the enhancer region. 5hmC, 5-hydroxymethylcytosine; 5mC, 5-methylcytosine; DHS, DNase hypersensitivity site; eRNA, enhancer RNA.

therefore do not formally rule out the possibility that a low level of H3K4me1 might be sufficient to maintain H3K27ac or that the enhancer state is maintained by the presence of an unidentified histone mark (or marks).

A clearer understanding of the sequence of events that takes place during enhancer activation has come from studying somatic cell reprogramming, which requires major epigenetic and transcriptional changes. During this cellular transition, transcription factor binding directs the establishment of *de novo* open chromatin regions and subsequent deposition of epigenetic marks. The initial binding sites of the reprogramming transcription factors octamer-binding protein 4 (OCT4; also known as PO5F1), Krüppel-like factor 4 (KLF4), SOX2 and MYC are mainly located at distal regulatory elements in the vicinity of silent genes^{136,137}, which need to be re-activated to re-establish the pluripotency network. Little evidence has been found for pre-existing active histone marks, such as H3K4me1, or for regions of accessible DNA that would guide these reprogramming transcription factors to their initial target sites^{101,102}. Therefore, it seems likely that changes in histone modification patterns occur after transcription factor binding¹³⁸. These observations imply that reprogramming factors need to be able to directly bind to closed chromatin regions to function as pioneer transcription factors and to initiate enhancer formation. How the reprogramming transcription factors find their target sites and which of the reprogramming transcription factors is first or key in binding to these regions remain open questions. The observation that early-activated enhancers tend to have lower DNA methylation levels¹³⁷ suggests that a less repressive chromatin structure permits transcription factor binding. By contrast, in somatic cells, regions of the genome with high levels of the repressive marks H3K9me2 or H3K9me3 seem refractory to binding of reprogramming and/or pioneer transcription factors^{136,139,140}. Accordingly, removing H3K9me3^{136,139,140} or reducing DNA methylation^{141,142} increases initial binding of the reprogramming transcription factors and improves the reprogramming efficiency.

Although enhancer priming has been observed in many cellular transitions, it seems not to be an obligatory step for enhancer activation¹⁴³. Analysis of enhancers in HSCs has revealed a dynamic and cell type-specific pattern in which ~40% of dynamic enhancers are established *de novo* during lineage commitment (that is, they are not present in HSCs) and become active in differentiated cells¹⁴⁴. For example, during B cell differentiation in mice, a high proportion of active enhancers seem to be established *de novo* as H3K4me1 or H3K27ac are not present in HSCs or in B cell progenitors¹⁴³.

In a study investigating the order of events during enhancer activation, combinatorial measurement of nascent RNAs, enhancer marks and transcription factor binding during macrophage activation showed that ~10% of enhancers gained *de novo* H3K4me2 marks after macrophage activation¹⁴⁵. Importantly, chromatin immunoprecipitation followed by sequencing (ChIP-seq; BOX 2) experiments indicated that transcription factor binding preceded H3K4me2 deposition at these sites. Inhibiting the binding of the transcription factor

nuclear factor- κ B (NF- κ B) resulted in reduced deposition of H3K4me2 and prevented *de novo* activation of these enhancers. Moreover, analysis of eRNA production revealed that enhancer transcription precedes H3K4 methylation in stimulated macrophages¹⁴⁵.

Together, these examples strongly suggest that transcription factor binding promotes a more accessible chromatin structure and precedes histone modifications in many cellular transitions. H3K4me1 seems to precede H3K27ac and thus to pre-mark a subset of enhancers for future activation during stem cell differentiation. However, this is not an obligatory step, as enhancers can be activated *de novo* in differentiated cells without prior H3K4me1 marking in progenitor cells.

H3K27me3, H3K4me3 and bivalency

Promoters of so-called bivalent genes are decorated both by activating H3K4me3 marks and by repressive H3K27me3 marks. Bivalency was first observed in mouse ESCs and sequential ChIP, and — more recently — the co-ChIP approach⁶⁴ (BOX 2) showed that both marks occurred on the same allele^{53,59,60,64,146,147}. Bivalency has also been detected in the developing embryo, indicating that the co-occurrence of H3K27me3 and H3K4me3 is not a peculiarity of *in vitro* culturing^{27,148,149}. According to the current model, co-occurrence of both activating and repressing histone marks safeguards against unscheduled perturbations in gene expression. In this scenario, H3K4me3 prevents the permanent silencing of genes, perhaps by preventing acquisition of DNA methylation, whereas H3K27me3 assures that gene expression levels remain low. As ESCs differentiate, bivalent loci lose one of the two histone marks and become fully activated or stably silenced. Emerging data indicate that bivalency is not restricted to stem cells and/or progenitor cells but that committed somatic cells can also acquire *de novo* bivalency, suggesting that this feature might provide dynamic gene responsiveness to signals, such as environmental cues, in somatic cells^{64,150,151}.

However, bivalency is not an essential feature for ESC pluripotency and self-renewal; ESCs cultured in 2i medium mostly lose promoter bivalency but maintain normal self-renewal and lineage differentiation potential^{12,76}. Consistent with these observations, low levels of promoter bivalency were found in pre-implantation embryos but sharply increased post-implantation^{27,28,57}. Accordingly, depleting mouse embryos of MLL2, the main writer of H3K4me3 at bivalent promoters, leads to growth retardation and embryonic lethality after gastrulation but has minimal effects on the self-renewal of cultured mouse ESCs^{152,153}. Similarly, depleting UTX (also known as KDM6A), an eraser of the H3K27me3 mark at bivalent promoters, leads to increased embryonic lethality in mice but does not affect mouse ESC self-renewal^{154–156}. Taken together, these findings suggest that bivalency is not an essential feature of naive pluripotent cells. However, promoter bivalency increases and becomes functionally more important during differentiation and at later developmental stages, such as the post-implantation embryo and the EpiSC state, in which cells do not tolerate the loss of H3K27me3.

DNA methylation and histone marks

The interplay between DNA methylation and deposition of H3K27me₃ has been extensively studied with the aim of resolving whether the two repressive marks have redundant, antagonizing or synergistic effects. Studies in ESCs and other cell types suggest that H3K27me₃ and DNA methylation are mutually exclusive at CpG islands⁵⁶. Moreover, it seems that DNA methylation can prevent the deposition of H3K27me₃ (REFS 56, 157) (FIG. 4). Indeed, many promoters in CpG islands bind TET proteins, which can maintain DNA methylation at low levels; depleting these TET proteins leads to increased DNA methylation and reduced H3K27me₃ at a large fraction of the CpG islands¹⁵⁸. DNA methylation is also negatively correlated with the H3K4me₃ mark^{159–161}. Thus, a loss or reduction

in TET activity leads to increased DNA methylation and consequently to loss of H3K4me₃ and H3K27me₃ — that is, loss of bivalency. Of note, the level of hydroxymethylation (for example, 5hmC) is low at CpG islands relative to other genomic regions¹⁶², raising the formal possibility that, instead of enzymatically oxidizing methylated DNA, TET binding might prevent binding of DNA methyltransferases at CpG islands; studies with catalytically inactive TET enzymes are required to investigate these bivalency-related epigenetic mechanisms.

Studies in ESCs with little or no DNA methylation, such as 2i-cultured ESCs or ESCs deficient for all three DNA methyltransferases, suggest that reduced DNA methylation leads to a spreading of H3K27me₃ from bivalent promoters into flanking genomic regions^{12,56}. In line with this observation, the PRC2 complex (the writer of H3K27me₃) was shown to have a lower affinity for methylated DNA than unmethylated DNA *in vitro*¹⁵⁷. Hence, removing DNA methylation may lead to the spread of H3K27me₃ to DNA that is otherwise protected from methylation. By contrast, depleting H3K27me₃ (by depleting the PRC2 complex) does not lead to a major increase in or spreading of DNA methylation. Instead, analysis of DNA methylation in H3K27me₃-depleted ESCs showed no overall gain of DNA methylation at bivalent promoters¹⁶³. Therefore, deposition of H3K27me₃ does not seem to prevent DNA methylation. It remains to be shown whether TET proteins are still enriched at bivalent sites and thus may antagonize DNA methylation even in the absence of H3K27me₃ and whether depleting TET proteins in PRC2-deficient cells increases DNA methylation at bivalent promoters.

In contrast to H3K27me₃, the H3K9me₂ mark is positively correlated with DNA methylation. The G9A–GLP complex is reported to recruit DNMT3A, DNMT3B and DNMT3L to maintain DNA methylation and genomic imprinting in mouse ESCs^{164–166}. H3K9me₂ is recognized by UHRF1, a component of the DNA methylation machinery, to facilitate DNA methylation following replication⁹⁶ (FIG. 4). Consistent with this role, loss of H3K9me₂ leads to global DNA demethylation in mouse ESCs^{107,164}, and ChIP-seq for H3K9me₂ followed by bisulfite sequencing revealed high DNA methylation at sites that have high levels of H3K9me₂ (REF. 107). H3K9me₂ is also recognized by STELLA (also known as DPPA3), which protects 5mC against conversion to 5hmC and consequently prevents DNA demethylation¹⁶⁷. This mechanism was first reported for maternal chromatin in early-stage zygotes, which harbour high H3K9me₂ levels that prevent DNA demethylation¹⁶⁷. Thus, H3K9me₂ is positively correlated with DNA methylation, and its reduction in PGCs and naive 2i-cultured ESCs is linked to global DNA demethylation.

DNA methylation is negatively correlated with chromatin accessibility¹⁶⁸. Bisulfite sequencing in mouse ESCs and neuronal progenitor cells identified regions with low levels (10–50%) of methylation that mainly overlapped with DHS sites¹⁶⁹. Depleting specific transcription factors resulted in increased DNA methylation at the corresponding binding sites, which suggests that binding of these transcription factors prevents the binding or

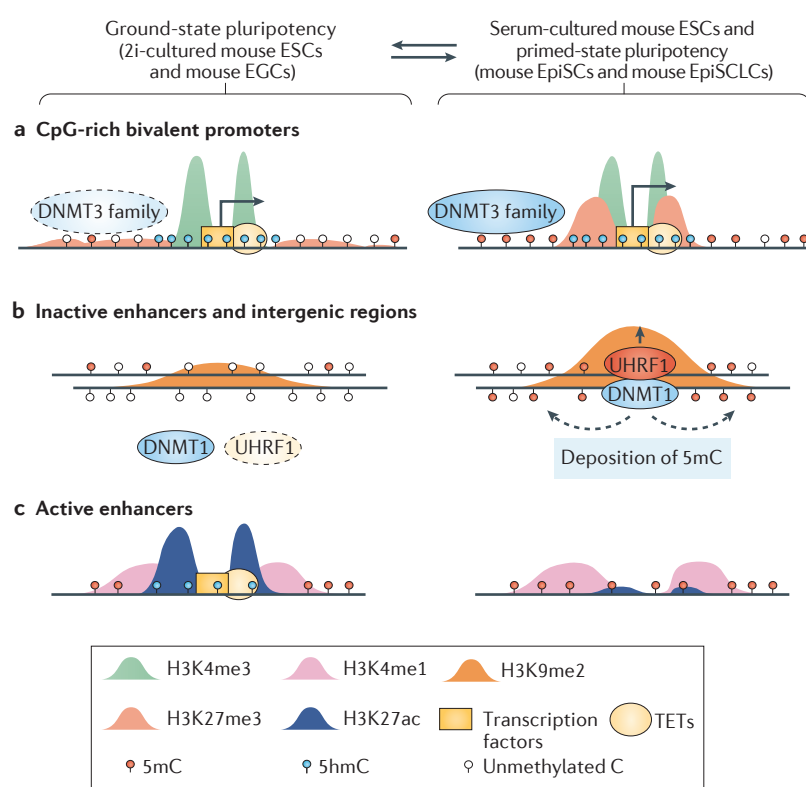


Figure 4 | The interplay between DNA methylation and histone modifications in different pluripotency states. **a** | In serum-cultured embryonic stem cells (ESCs) and epiblast-derived stem cells (EpiSCs; right), a high level of 5-methylcytosine (5mC) DNA methylation restricts trimethylation of histone H3 at lysine 27 (H3K27me₃) deposition to CpG-rich bivalent promoters. In ground-state ESCs (left), global DNA demethylation leads to carpeting (widespread but at low levels) of the genome by H3K27me₃ and reduced H3K27me₃ deposition at bivalent promoters. The activity of TET proteins ensures high level of 5-hydroxymethylcytosine (5hmC) at these promoters. **b** | The lower level of global DNA methylation (including at enhancers and intergenic regions) in ground-state pluripotency is a result of: reduced levels of DNA methyltransferase 3A (DNMT3A), DNMT3B and DNMT3L (required for *de novo* DNA methylation); and, in particular, reduced levels of UHRF1 and H3K9me₂ (left), which leads to impairment of DNMT1 function (required for maintenance of DNA methylation). Reduced DNMT1 activity or UHRF1 levels prevents restoration of full DNA methylation at hemimethylated sites generated at replication forks and leads to global DNA demethylation. **c** | DNA methylation (5mC) is minimal at enhancers at which transcription factor binding and TET activity prevent DNA methylation, enabling deposition of H3K27ac and H3K4me₁ and ensuring high levels of 5hmC at enhancers. ESCs, embryonic germ cells.

activity of the DNA methylation machinery and results in low levels of DNA methylation at regulatory elements¹⁶⁹. High levels of DNA methylation and low chromatin accessibility at enhancers are generally linked to loss of activating histone marks, such as H3K27ac. However, what seemed to be a firm rule was challenged by observations that DNA methylation and the H3K27ac mark coexist at enhancer regions in both tumour cell lines and normal cells¹⁷⁰. In colon cancer cells, 64% of acetylated super-enhancers and 38% of acetylated regular enhancers have high levels (more than 70%) of DNA methylation. Intriguingly, deposition of H3K27ac at these enhancers was dependent on DNA methylation, as treatment with an inhibitor of DNA methylation resulted in loss of the H3K27ac mark¹⁷⁰. This study suggests an active role for DNA methylation in maintaining enhancer acetylation. Whether these observations originate from cellular heterogeneity rather than true bivalency at these enhancer loci remains to be determined.

Taken together, DNA methylation shows a dynamic pattern in different ESC states and functionally interacts with other epigenetic marks: it prevents deposition of H3K27me3, functions in synergy with H3K9me2 and H3K9me3, and influences transcription factor binding, chromatin accessibility and enhancer activity. It is worth mentioning that ESCs can tolerate the complete loss of DNA methylation, whereas this type of modification becomes indispensable in EpiSCs and later developmental stages^{71,171,172}. Whether inducing high levels of DNA methylation in naive ESCs (such as 2i-cultured ESCs) affects their self-renewal and differentiation remains to be examined.

Chromatin structure and epigenetic marks

How higher-order chromatin organization (such as TAD formation) and lower-order chromatin loops (for example, enhancer–promoter interactions) are established and whether this process is initiated by transcription factors and stabilized by histone modifications are areas of intense investigation.

Hi-C analysis of the chromatin organization in human ESCs and their differentiated progenies suggest that H3K4me1 is a predictor of chromatin loop interactions¹¹². Accordingly, depleting the MLL3–MLL4 complex in mouse ESCs reduced the level of H3K4me1 at some enhancers and resulted in reduced enhancer–promoter looping³⁹. Thus, these studies suggest a role for H3K4me1 in establishing dynamic enhancer–promoter looping in addition to its role as a general enhancer mark; however, the extent to which H3K4me1 can establish chromatin loop interactions requires further investigation. In another study, epigenetic modifications and transcription factor-binding profiles were linked to Hi-C interaction maps to generate a complex epigenetic signature that could predict physical interactions between enhancers and promoters¹⁷³. By comparison, histone marks (including H3K4me1), eRNAs, chromatin accessibility, structural proteins and different transcription factors were shown to have low predictive values when used individually¹⁷³.

One prominent feature of chromatin organization that has been shown to be mediated by epigenetic marks, in

this case H3K27me3 (REFS 174–176), is extreme long-range interactions (ELRI) between promoters. ELRIs, which extend over hundreds of kilobases in mouse ESCs, are formed between a subset of H3K27me3-decorated promoters (including the *HOX* loci) in mouse ESCs cultured in serum-supplemented medium¹⁷⁵. However, in ground-state pluripotency (2i-cultured mouse ESCs), H3K27me3 is not localized but spread throughout the genome^{12,175} and is probably the cause of ELRI dismantling during the transition from serum to 2i-cultured mouse ESC states. Consistent with this hypothesis, long-range promoter–promoter interactions are lost in mouse ESCs depleted for components of the PRC1 or PRC2 complexes, suggesting that colocalization of both of these complexes is necessary but possibly not sufficient to establish and/or to maintain such long-range interactions^{174,175}. Indeed, the presence of H3K27me3, PRC1 and PRC2 does not appear to be the sole determinant of ELRI, as many loci decorated with this repressive mark and bound by PRC1 and PRC2 factors are not involved in ELRI^{174,175}. Therefore, specialized PRC1 and PRC2 complexes might be required. Interestingly, PRC2 was also found to mediate long-range interactions in human embryonic carcinoma cells, and depleting the PRC2 component EZH2 reduced these interactions¹⁷⁷. In line with these observations, a direct role for the PRC2 complex in establishing new chromatin contacts was shown using EZH2-tethering experiments: targeting EZH2 to specific genomic loci resulted in the formation of new chromatin contacts and was accompanied by the deposition of H3K27me3. However, these newly generated loops did not affect the expression of genes located in the surrounding chromatin¹⁷⁸. ELRIs therefore represent dynamic chromatin interactions (promoter–promoter interactions) that are mediated by H3K27me3 (and PRC1 and PRC2) and are established in different ESC states, but the biological importance of these interactions and whether they exist *in vivo* remains unknown.

Histone modifications might also be involved in higher-order nuclear organization by regulating nuclear lamina-associated domains (LADs) (FIG. 2a). LADs are heterochromatic regions decorated with H3K27me3 and H3K9me2 and contain genes that are mainly (but not exclusively) silent^{179,180}. Repositioning of heterochromatin to the nuclear lamina has been observed in several cellular transitions, including ESC differentiation¹⁸¹, and may provide a mechanism of repression. In this regard, alterations in histone modifications (such as H3K9me2) might be linked to changes in nuclear architecture and might contribute to gene silencing by repositioning genes close to the nuclear periphery^{29,77,182}. Such an occurrence has been observed during *in vitro* differentiation of mouse PGCs, in which dense spots of lamin A and lamin B1 and the heterochromatin around the nuclear membrane are reduced at the same time as the genes necessary for PGC differentiation are activated²⁹.

In addition to histone modifications, transcription factor binding also seems to play an important, maybe even essential, part in establishing chromatin loops. A large body of evidence suggests that TAD organization, as well as the formation of a substantial number of enhancer–promoter loops, are mediated by CTCF–cohesin

Super-enhancers

Densely clustered enhancers that control genes involved in regulating the cell identity.

Hi-C

A chromosome conformation capture technique that maps all the chromatin interactions in the genome.

Extreme long-range interactions (ELRI)

Promoter–promoter interactions that occur over megabase distances and are associated with trimethylation of histone H3 at lysine 27 (H3K27me3) marks in embryonic stem cells (ESCs).

Ground-state pluripotency

A state of pluripotency detected in pre-implantation epiblast cells and in 2i-cultured embryonic stem cells (ESCs) that is independent of many repressive epigenetic marks.

CRISPR–Cas9 editing technology

A targeted genome-editing tool that has been engineered from components of a prokaryotic immune system.

complexes^{113,122,183,184}, which raises the question whether DNA and histone modifications can affect CTCF and/or cohesin binding and consequently alter chromatin interactions. In human somatic cells, CTCF binding was shown to be sensitive to DNA methylation, in which increased DNA methylation reduces CTCF binding and chromatin looping^{185,186}. These observations suggest that DNA methylation might directly or indirectly influence enhancer–promoter looping and chromatin organization. The effect of DNA methylation on CTCF binding in ESCs requires further investigation¹⁶⁹.

In summary, epigenetic modifications can directly reshape the 3D genome in several ways, including facilitating enhancer–promoter or promoter–promoter interactions, altering nuclear localization of chromatin or influencing binding of transcription factors and architectural proteins that mediate chromatin interactions.

Perspectives and conclusions

Our understanding of the role of epigenetics in cellular homeostasis and disease states has substantially increased since the term was first coined by Waddington in 1942. It has become clear that the different epigenetic mechanisms are closely interconnected and often function together to control expression of the genetic code. However, more integrative approaches are required to connect the different layers of epigenetic information in a spatiotemporal manner. Time-resolved studies of cellular transitions or responses to external stimuli will provide information on the hierarchy of epigenetic marks and events that cannot be obtained by investigating steady states. Such studies might also reveal whether epigenetic modifications are directive, supportive or a consequence of transcription factor-instigated events. In this regard, emerging CRISPR–Cas9 editing technology¹⁸⁷ will permit targeted manipulation of epigenetic mechanisms to elucidate the causal role, if any, of epigenetic marks in a locus-specific manner.

Recent advances in understanding chromatin organization have provided new insights into the principles of gene regulation but have not yet led to a deeper understanding of how promoter–promoter, enhancer–promoter and enhancer–enhancer networks are established and dismantled during different cellular transitions. Similarly, it is poorly understood to what extent dynamic chromatin interactions direct or facilitate differential gene expression. As the TAD structure provides an architectural framework that is mostly stable between different cellular states, further studies are needed to reveal whether enhancer–promoter interactions form ‘functional’ interaction hubs that are more dynamic than TADs and that can better explain the highly dynamic transcriptional response.

A unifying picture has emerged in which (pioneer) transcription factors initiate enhancer activation by

binding to *cis*-regulatory elements embedded in poorly accessible chromatin (FIG. 3). These transcription factors in turn recruit chromatin-remodelling complexes and facilitate the binding of reader, eraser and writer complexes. These proteins then remove the repressive marks, deposit active marks and ultimately activate enhancers. Conversely, enhancer decommissioning is probably initiated by the loss of transcription factor binding, leading to epigenetic instability, removal of active marks, restriction of chromatin accessibility and ‘sealing’ the locus with repressive marks^{14,36,40,96,128–130,133–135,169,188}. However, whether there is a fixed temporal order to these events remains to be addressed. In addition, the potential synergy, redundancy or antagonism between enhancers that dictates the final transcriptional output of a target gene is poorly understood. New genome-wide methods that examine enhancer function in a high-throughput manner¹⁸⁹ might help to elucidate the relative contribution of different enhancers and to determine whether some enhancers have a crucial role in controlling the enhancer–promoter interaction network and, ultimately, gene expression.

In recent years, new connections have emerged between epigenetics and cell mechanisms that were seemingly independent of epigenetic regulation². For instance, there is an accumulating body of evidence indicating that DNA and histone modifications are closely linked to cellular metabolism, in which different metabolites function as substrate donors or inhibitors of chromatin modifications¹⁹⁰. We anticipate major advances in our understanding of how cellular metabolism alters the epigenome, 3D genome organization and gene expression. This research is also predicted to reveal new insights into how environmental factors (such as diet) influence cellular metabolism and epigenetic modifications. It is important to mention that our understanding of epigenetics is so far based on the few histone marks that have been studied in sufficient detail. We have only uncovered the ‘tip of iceberg’ of epigenetic regulation. Moreover, most epigenetic studies have been carried out in (cancer) cell lines and therefore have to be interpreted with caution. There has only recently been a shift towards the use of primary cells. The next challenge will be to investigate epigenetic changes in the context of complex tissues (and their *in vitro* organoid models), in which interactive cell–cell contacts and signalling pathways take place. Lastly, as the vast majority of current epigenetic profiles and 3D chromatin organization data are based on averaging cell populations, low-input and single-cell methods¹⁹¹ will substantially sharpen and expand our view of the epigenetic heterogeneity and chromatin organization across cells. Combined with high-resolution microscopy¹⁹², methods for detecting chromatin interactions and epigenetic marking will provide better insights into epigenetic regulation *in vivo* in the developing embryo or during adult stem cell differentiation.

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